

РБНФ №1 (опис синтаксису всіма допустимими засобами РБНФ)	РБНФ №2 (опис формальної граматики засобами РБНФ)	Формальна граматика	Формальна граматика з специфікацією lookahead у правилах для LL(2)-аналізатора	Перевірка РБНФ №1 за допомогою коду (помістити у файл "EBNF_N1.h") */	Перевірка РБНФ №2 за допомогою коду (помістити у файл "EBNF_N2.h") */	Перевірка прототипу LL(2)-синтаксичного аналізатора (спеціальна структура) та прототипу лексичного аналізатора (регулярні вирази) за допомогою коду. Лексеми для синтаксичного аналізатора обробляються лексичним аналізатором, тому синтаксичний аналізатор не аналізує їх повторно (як показано в РБНФ). (помістити у файл "LexicaByRegExAndSyntaxByLL2prototype.h") УВАГА: при копіюванні заважайте, щоб у кожному рядку після символу «\» не містилось жодних інших символів. */
		G = (N, T, P, S)	G = (N, T, P, S)			
		S → program_rule	S → program_rule			
		N = { labeled_point, goto_label, program_name, value_type, array_specify, declaration_element, array_specify_optional, other_declaration_ident, declaration, other_declaration_ident_iteration, index_action, unary_operator, unary_operation, binary_operator, binary_action, left_expression, group_expression, index_action_optional, expression, binary_action_iteration, expression_or_cond_block_with_optional_assign, assign_to_right, assign_to_right_optional, if_expression, body_for_true, false_cond_block_without_else, body_for_false, cond_block, false_cond_block_without_else_iteration, body_for_false_optional, statement, block_statements, input_rule, argument_for_input, output_rule, statement_iteration, expression_optional, program_rule, declaration_optional, non_zero_digit, digit_iteration, digit, unsigned_value, value, sign_optional, sign, ident, letter_in_upper_case, letter_in_lower_case, sign_plus, sign_minus }	N = { labeled_point, goto_label, program_name, value_type, array_specify, declaration_element, array_specify_optional, other_declaration_ident, declaration, other_declaration_ident_iteration, index_action, unary_operator, unary_operation, binary_operator, binary_action, left_expression, group_expression, index_action_optional, expression, binary_action_iteration, expression_or_cond_block_with_optional_assign, assign_to_right, assign_to_right_optional, if_expression, body_for_true, false_cond_block_without_else, body_for_false, cond_block, false_cond_block_without_else_iteration, body_for_false_optional, statement, block_statements, input_rule, argument_for_input, output_rule, statement_iteration, expression_optional, program_rule, declaration_optional, non_zero_digit, digit_iteration, digit, unsigned_value, value, sign_optional, sign, ident, letter_in_upper_case, letter_in_lower_case, sign_plus, sign_minus }	#define NONTERMINALS labeled_point, \ goto_label, \ program_name, \ value_type, \ array_specify, \ declaration_element, \ \ other_declaration_ident, \ declaration, \ \ index_action, \ unary_operator, \ unary_operation, \ binary_operator, \ binary_action, \ left_expression, \ group_expression, \ \ expression, \ \ expression_or_cond_block_with_optional_assign, \ assign_to_right, \ \ if_expression, \ body_for_true, \ false_cond_block_without_else, \ body_for_false, \ cond_block, \ \ statement, \ input_rule, \ argument_for_input, \ output_rule, \ \ program_rule, \ \ non_zero_digit, \ digit_iteration, \ digit, \ unsigned_value, \ value, \ \ sign, \ ident, \ letter_in_upper_case, \ letter_in_lower_case, \ sign_plus, \ sign_minus	#define NONTERMINALS labeled_point, \ goto_label, \ program_name, \ value_type, \ array_specify, \ declaration_element, \ array_specify_optional, \ other_declaration_ident, \ declaration, \ other_declaration_ident_iteration, \ index_action, \ unary_operator, \ unary_operation, \ binary_operator, \ binary_action, \ left_expression, \ group_expression, \ index_action_optional, \ expression, \ binary_action_iteration, \ expression_or_cond_block_with_optional_assign, \ assign_to_right, \ assign_to_right_optional, \ if_expression, \ body_for_true, \ false_cond_block_without_else, \ body_for_false, \ cond_block, \ false_cond_block_without_else_iteration, \ body_for_false_optional, \ statement, \ block_statements, \ input_rule, \ argument_for_input, \ output_rule, \ statement_iteration, \ expression_optional, \ program_rule, \ declaration_optional, \ non_zero_digit, \ digit_iteration, \ digit, \ unsigned_value, \ value, \ sign_optional, \ sign, \ ident, \ letter_in_upper_case, \ letter_in_lower_case, \ sign_plus, \ sign_minus	
		T = { ",", ", ", "GOTO", "INTEGER16", ",", ", "NOT", "AND", "OR", "=", "!=", "<=", ">=", "+", ",", "=", "DIV", "MOD", "(", ")", "=", "ELSE", "IF",	T = { ",", ", ", "GOTO", "INTEGER16", ",", ", "NOT", "AND", "OR", "=", "!=", "<=", ">=", "+", ",", "=", "DIV", "MOD", "(", ")", "=", "ELSE", "IF",	#define TOKENS \ tokenCOLON, \ tokenGOTO, \ tokenINTEGER16, \ tokenCOMMA, \ tokenNOT, \ tokenAND, \ tokenOR, \ tokenEQUAL, \ tokenNOTEQUAL, \ tokenLESSOREQUAL, \ tokenGREATEROREQUAL, \ tokenPLUS, \ tokenMINUS, \ tokenMUL, \ tokenDIV, \ tokenMOD, \ tokenGROUPEXPRESSIONBEGIN, \ tokenGROUPEXPRESSIONEND, \ tokenLRASSIGN, \ tokenELSE, \ tokenIF, \	#define TOKENS \ tokenCOLON, \ tokenGOTO, \ tokenINTEGER16, \ tokenCOMMA, \ tokenNOT, \ tokenAND, \ tokenOR, \ tokenEQUAL, \ tokenNOTEQUAL, \ tokenLESSOREQUAL, \ tokenGREATEROREQUAL, \ tokenPLUS, \ tokenMINUS, \ tokenMUL, \ tokenDIV, \ tokenMOD, \ tokenGROUPEXPRESSIONBEGIN, \ tokenGROUPEXPRESSIONEND, \ tokenLRASSIGN, \ tokenELSE, \ tokenIF, \	

				tokenSEMICOLON = ";;" >> BOUNDARIES;	tokenSEMICOLON = ";;" >> BOUNDARIES;	#define T_SEMICOLON_0 ";" #define T_SEMICOLON_1 "" #define T_SEMICOLON_2 "" #define T_SEMICOLON_3 ""
				tokenCOLON = ":" >> BOUNDARIES;	tokenCOLON = ":" >> BOUNDARIES;	#define T_COLON_0 ":" #define T_COLON_1 "" #define T_COLON_2 "" #define T_COLON_3 ""
				tokenGOTO = "GOTO" >> STRICT_BOUNDARIES;	tokenGOTO = "GOTO" >> STRICT_BOUNDARIES;	#define T_GOTO_0 "GOTO" #define T_GOTO_1 "" #define T_GOTO_2 "" #define T_GOTO_3 ""
				tokenINTEGER16 = "INTEGER16" >> STRICT_BOUNDARIES;	tokenINTEGER16 = "INTEGER16" >> STRICT_BOUNDARIES;	#define T_DATA_TYPE_0 "INTEGER16" #define T_DATA_TYPE_1 "" #define T_DATA_TYPE_2 "" #define T_DATA_TYPE_3 ""
				tokenCOMMA = "," >> BOUNDARIES;	tokenCOMMA = "," >> BOUNDARIES;	#define T_COMA_0 "," #define T_COMA_1 "" #define T_COMA_2 "" #define T_COMA_3 ""
						#define T_BITWISE_NOT_0 "~" #define T_BITWISE_NOT_1 "" #define T_BITWISE_NOT_2 "" #define T_BITWISE_NOT_3 ""
				tokenNOT = "NOT" >> STRICT_BOUNDARIES;	tokenNOT = "NOT" >> STRICT_BOUNDARIES;	#define T_NOT_0 "NOT" #define T_NOT_1 "" #define T_NOT_2 "" #define T_NOT_3 ""
						#define T_BITWISE_AND_0 "&" #define T_BITWISE_AND_1 "" #define T_BITWISE_AND_2 "" #define T_BITWISE_AND_3 ""
				tokenAND = "AND" >> STRICT_BOUNDARIES;	tokenAND = "AND" >> STRICT_BOUNDARIES;	#define T_AND_0 "AND" #define T_AND_1 "" #define T_AND_2 "" #define T_AND_3 ""
						#define T_BITWISE_OR_0 " " #define T_BITWISE_OR_1 "" #define T_BITWISE_OR_2 "" #define T_BITWISE_OR_3 ""
				tokenOR = "OR" >> STRICT_BOUNDARIES;	tokenOR = "OR" >> STRICT_BOUNDARIES;	#define T_OR_0 "OR" #define T_OR_1 "" #define T_OR_2 "" #define T_OR_3 ""
				tokenEQUAL = "==" >> BOUNDARIES;	tokenEQUAL = "==" >> BOUNDARIES;	#define T_EQUAL_0 "==" #define T_EQUAL_1 "" #define T_EQUAL_2 "" #define T_EQUAL_3 ""
				tokenNOTEQUAL = "!=" >> BOUNDARIES;	tokenNOTEQUAL = "!=" >> BOUNDARIES;	#define T_NOT_EQUAL_0 "!=" #define T_NOT_EQUAL_1 "" #define T_NOT_EQUAL_2 "" #define T_NOT_EQUAL_3 ""
				tokenLESSOREQUAL = "<=" >> BOUNDARIES;	tokenLESSOREQUAL = "<=" >> BOUNDARIES;	#define T_LESS_OR_EQUAL_0 "<="
				tokenGREATEROREQUAL = ">=" >> BOUNDARIES;	tokenGREATEROREQUAL = ">=" >> BOUNDARIES;	#define T_GREATER_OR_EQUAL_0 ">="
				tokenPLUS = "+" >> BOUNDARIES;	tokenPLUS = "+" >> BOUNDARIES;	#define T_ADD_0 "+" #define T_ADD_1 "" #define T_ADD_2 "" #define T_ADD_3 ""
				tokenMINUS = "-" >> BOUNDARIES;	tokenMINUS = "-" >> BOUNDARIES;	#define T_SUB_0 "-" #define T_SUB_1 "" #define T_SUB_2 "" #define T_SUB_3 ""
				tokenMUL = "*" >> BOUNDARIES;	tokenMUL = "*" >> BOUNDARIES;	#define T_MUL_0 "" #define T_MUL_1 "" #define T_MUL_2 "" #define T_MUL_3 ""
				tokenDIV = "DIV" >> STRICT_BOUNDARIES;	tokenDIV = "DIV" >> STRICT_BOUNDARIES;	#define T_DIV_0 "DIV" #define T_DIV_1 "" #define T_DIV_2 "" #define T_DIV_3 ""
				tokenMOD = "MOD" >> STRICT_BOUNDARIES;	tokenMOD = "MOD" >> STRICT_BOUNDARIES;	#define T_MOD_0 "MOD" #define T_MOD_1 "" #define T_MOD_2 "" #define T_MOD_3 ""
				tokenLRASSIGN = "=:;" >> BOUNDARIES;	tokenLRASSIGN = "=:;" >> BOUNDARIES;	#define T_LRASSIGN_0 "=:;" #define T_LRASSIGN_1 "" #define T_LRASSIGN_2 "" #define T_LRASSIGN_3 ""
						#define T_THEN_BLOCK_0 "{" #define T_THEN_BLOCK_1 "" #define T_THEN_BLOCK_2 "" #define T_THEN_BLOCK_3 ""
				tokenELSE = "ELSE" >> STRICT_BOUNDARIES;	tokenELSE = "ELSE" >> STRICT_BOUNDARIES;	#define T_ELSE_BLOCK_0 "ELSE" #define T_ELSE_BLOCK_1 T_BEGIN_BLOCK_0 #define T_ELSE_BLOCK_2 "" #define T_ELSE_BLOCK_3 ""
				tokenIF = "IF" >> STRICT_BOUNDARIES;	tokenIF = "IF" >> STRICT_BOUNDARIES;	#define T_IF_0 "IF" #define T_IF_1 "" #define T_IF_2 "" #define T_IF_3 ""
						#define T_ELSE_IF_0 "ELSE" #define T_ELSE_IF_1 T_IF_0 #define T_ELSE_IF_2 "" #define T_ELSE_IF_3 ""
				tokenEXIT = "EXIT" >> STRICT_BOUNDARIES;	tokenEXIT = "EXIT" >> STRICT_BOUNDARIES;	#define T_EXIT_0 "EXIT" #define T_EXIT_1 "" #define T_EXIT_2 "" #define T_EXIT_3 ""
				tokenGET = "GET" >> STRICT_BOUNDARIES;	tokenGET = "GET" >> STRICT_BOUNDARIES;	#define T_INPUT_0 "GET" #define T_INPUT_1 "" #define T_INPUT_2 ""

						#define T_INPUT_3 ""
				tokenPUT = "PUT" >> STRICT_BOUNDARIES;	tokenPUT = "PUT" >> STRICT_BOUNDARIES;	#define T_OUTPUT_0 "PUT" #define T_OUTPUT_1 "" #define T_OUTPUT_2 "" #define T_OUTPUT_3 ""
				tokenNAME = "NAME" >> STRICT_BOUNDARIES;	tokenNAME = "NAME" >> STRICT_BOUNDARIES;	#define T_NAME_0 "NAME" #define T_NAME_1 "" #define T_NAME_2 "" #define T_NAME_3 ""
				tokenBODY = "BODY" >> STRICT_BOUNDARIES;	tokenBODY = "BODY" >> STRICT_BOUNDARIES;	#define T_BODY_0 "BODY" #define T_BODY_1 "" #define T_BODY_2 "" #define T_BODY_3 ""
				tokenDATA = "DATA" >> STRICT_BOUNDARIES;	tokenDATA = "DATA" >> STRICT_BOUNDARIES;	#define T_DATA_0 "DATA" #define T_DATA_1 "" #define T_DATA_2 "" #define T_DATA_3 ""
				tokenBEGIN = "BEGIN" >> STRICT_BOUNDARIES;	tokenBEGIN = "BEGIN" >> STRICT_BOUNDARIES;	#define T_BEGIN_0 "BEGIN" #define T_BEGIN_1 "" #define T_BEGIN_2 "" #define T_BEGIN_3 ""
				tokenEND = "END" >> STRICT_BOUNDARIES;	tokenEND = "END" >> STRICT_BOUNDARIES;	#define T_END_0 "END" #define T_END_1 "" #define T_END_2 "" #define T_END_3 ""
						#define T_NULL_STATEMENT_0 "NULL" #define T_NULL_STATEMENT_1 "STATEMENT" #define T_NULL_STATEMENT_2 "" #define T_NULL_STATEMENT_3 ""
						#define GRAMMAR_LL2_2025 {\
labeled_point = ident , ":";	labeled_point = ident , ":";	labeled_point → ident ":"	labeled_point(1: "ident_terminal") → ident ":"	labeled_point = ident >> tokenCOLON;	labeled_point = ident >> tokenCOLON;	{ LA_IS, {"ident_terminal"}, {"labeled_point"}, {\
goto_label = "GOTO" , ident;	goto_label = "GOTO" , ident;	goto_label → "GOTO" ident	goto_label(1: "GOTO") → "GOTO" ident	goto_label = tokenGOTO >> ident;	goto_label = tokenGOTO >> ident;	{ LA_IS, {T_GOTO_0}, {"goto_label"}, {\
program_name = ident;	program_name = ident;	program_name → ident	program_name(1: "ident_terminal") → ident	program_name = SAME_RULE(ident);	program_name = SAME_RULE(ident);	{ LA_IS, {"ident_terminal"}, {"program_name"}, {\
value_type = "INTEGER16";	value_type = "INTEGER16";	value_type → "INTEGER16"	value_type(1: "INTEGER16") → "INTEGER16"	value_type = SAME_RULE(tokenINTEGER16);	value_type = SAME_RULE(tokenINTEGER16);	{ LA_IS, {T_DATA_TYPE_0}, {"value_type"}, {\
	array_specify = "[" , unsigned_value , "]" ;	array_specify → "[" unsigned_value "]"	array_specify(1: "[") → "[" unsigned_value "]"		array_specify = "[" >> unsigned_value >> "]" ;	{ LA_IS, {"["}, {"array_specify"}, {\
declaration_element = ident , ["[" , unsigned_value , "]"] ;	declaration_element = ident , array_specify__optional;	declaration_element → ident array_specify__optional	declaration_element(1: "ident_terminal") → ident array_specify__optional	declaration_element = ident >> -(tokenLEFTSQUAREBRACKETS >> unsigned_value >> tokenRIGHTSQUAREBRACKETS);	declaration_element = ident >> array_specify__optional;	{ LA_IS, {"ident_terminal"}, {"declaration_element"}, {\
	array_specify__optional = array_specify ε;	array_specify__optional → array_specify array_specify__optional → ε	array_specify__optional(1: "[") → array_specify array_specify__optional(1: !"[") → ε		array_specify__optional = array_specify "";	{ LA_IS, {"["}, {"array_specify__optional"}, {\
other_declaration_ident = " , " , declaration_element;	other_declaration_ident = " , " , declaration_element;	other_declaration_ident → " , " , declaration_element	other_declaration_ident(1: " , ") → " , " , declaration_element	other_declaration_ident = tokenCOMMA >> declaration_element;	other_declaration_ident = tokenCOMMA >> declaration_element;	{ LA_IS, {T_COMA_0}, {"other_declaration_ident"}, {\
declaration = value_type , declaration_element , {other_declaration_ident};	declaration = value_type , declaration_element , other_declaration_ident__iteration;	declaration → value_type declaration_element other_declaration_ident__iteration	declaration(1: "INTEGER16") → value_type declaration_element other_declaration_ident__iteration	declaration = value_type >> declaration_element >> *other_declaration_ident;	declaration = value_type >> declaration_element >> other_declaration_ident__iteration;	{ LA_IS, {T_DATA_TYPE_0}, {"declaration"}, {\
	other_declaration_ident__iteration = other_declaration_ident , other_declaration_ident__iteration ε;	other_declaration_ident__iteration → other_declaration_ident other_declaration_ident__iteration false_cond_block_without_else__iteration → ε	other_declaration_ident__iteration(1: " , ") → other_declaration_ident other_declaration_ident__iteration false_cond_block_without_else__iteration(1: !" , ") → ε		other_declaration_ident__iteration = other_declaration_ident >> other_declaration_ident__iteration "";	{ LA_IS, {T_COMA_0}, {\
index_action = "[" , expression , "]" ;	index_action = "[" , expression , "]" ;	index_action → "[" expression "]"	index_action(1: "[") → "[" expression "]"	index_action = tokenLEFTSQUAREBRACKETS >> expression >> tokenRIGHTSQUAREBRACKETS;	index_action = tokenLEFTSQUAREBRACKETS >> expression >> tokenRIGHTSQUAREBRACKETS;	{ LA_IS, {"["}, {"index_action"}, {\
unary_operator = "NOT";	unary_operator = "NOT";	unary_operator → "NOT"	unary_operator(1: "NOT") → "NOT"	unary_operator = SAME_RULE(tokenNOT);	unary_operator = SAME_RULE(tokenNOT);	{ LA_IS, {T_NOT_0}, {"unary_operator"}, {\
unary_operation = unary_operator , expression;	unary_operation = unary_operator , expression;	unary_operation → unary_operator expression	unary_operation(1: "NOT") → unary_operator expression	unary_operation = unary_operator >> expression;	unary_operation = unary_operator >> expression;	{ LA_IS, {T_NOT_0}, {"unary_operation"}, {\
binary_operator = "AND" "OR" "==" " =" "<=" ">=" "+" "-" "*" "DIV" "MOD";	binary_operator = "AND" "OR" "==" " =" "<=" ">=" "+" "-" "*" "DIV" "MOD";	binary_operator → "AND" binary_operator → "OR" binary_operator → "==" binary_operator → " =" binary_operator → "<=" binary_operator → ">=" binary_operator → "+" binary_operator → "-" binary_operator → "*" binary_operator → "DIV" binary_operator → "MOD"	binary_operator(1: "AND") → "AND" binary_operator(1: "OR") → "OR" binary_operator(1: "==") → "==" binary_operator(1: " =") → " =" binary_operator(1: "<=") → "<=" binary_operator(1: ">=") → ">=" binary_operator(1: "+") → "+" binary_operator(1: "-") → "-" binary_operator(1: "*") → "*" binary_operator(1: "DIV") → "DIV" binary_operator(1: "MOD") → "MOD"	binary_operator = tokenAND tokenOR tokenEQUAL tokenNOTEQUAL tokenLESSOREQUAL tokenGREATEROREQUAL tokenPLUS tokenMINUS tokenMUL tokenDIV tokenMOD;	binary_operator = tokenAND tokenOR tokenEQUAL tokenNOTEQUAL tokenLESSOREQUAL tokenGREATEROREQUAL tokenPLUS tokenMINUS tokenMUL tokenDIV tokenMOD;	{ LA_IS, {T_AND_0}, {"binary_operator"}, {\

	false_cond_block_without_else__iteration ε;	false_cond_block_without_else__iteration false_cond_block_without_else__iteration → ε	false_cond_block_without_else__iteration false_cond_block_without_else__iteration(1: "ELSE"; 2: !"IF") → ε false_cond_block_without_else__iteration(1: !"ELSE") → ε		false_cond_block_without_else__iteration "";	{LA_IS, {T_ELSE_IF_1}, 2, { "false_cond_block_without_else", "false_cond_block_without_else__iteration" }}\} {LA_NOT, {T_ELSE_IF_1}, 0, { "" }}\} }}\} {LA_NOT, {T_ELSE_IF_0}, { "false_cond_block_without_else__iteration", {\ {LA_IS, {""}, 0, { "" }}\} }}\}
	body_for_false__optional = body_for_false ε;	body_for_false__optional → body_for_false body_for_false__optional → ε	body_for_false__optional(1: "FALSE") → body_for_false body_for_false__optional(1: !"FALSE") → ε		body_for_false__optional = body_for_false "";	{LA_IS, {T_ELSE_BLOCK_0}, { "body_for_false__optional", {\ {LA_IS, {""}, 1, { "body_for_false" }}\} }}\} {LA_NOT, {T_ELSE_BLOCK_0}, { "body_for_false__optional", {\ {LA_IS, {""}, 0, { "" }}\} }}\}
input_rule = "GET" , (ident , [index_action] "(" , ident , [index_action] , ")");	input_rule = "GET", argument_for_input;	input_rule → "GET" argument_for_input	input_rule(1: "GET") → "GET" argument_for_input	input_rule = tokenGET >> {ident >> -index_action tokenGROUPEXPRESSIONBEGIN >> ident >> -index_action >> tokenGROUPEXPRESSIONEND};	input_rule = tokenGET >> argument_for_input;	{LA_IS, {T_INPUT_0}, { "input_rule", {\ {LA_IS, {""}, 2, { T_INPUT_0, "argument_for_input" }}\} }}\}
	argument_for_input = ident , index_action__optional; argument_for_input = "(" , "ident" , "index_action__optional" , ")";	argument_for_input → ident index_action__optional argument_for_input → "(" (" ident" "index_action__optional" ")"	argument_for_input(1: " ") → ident index_action__optional argument_for_input(1: "(") → "(" (" ident" "index_action__optional" ")"		argument_for_input = ident >> index_action__optional tokenGROUPEXPRESSIONBEGIN >> ident >> index_action__optional >> tokenGROUPEXPRESSIONEND;	{LA_IS, { " ident_terminal " }, { "argument_for_input", {\ {LA_IS, {""}, 2, { "ident", "index_action__optional" }}\} }}\} {LA_IS, { "(" }, { "argument_for_input", {\ {LA_IS, {""}, 4, { "(" , "ident", "index_action__optional", ")" }}\} }}\}
output_rule = "PUT" , expression;	output_rule = "PUT", expression;	output_rule → "PUT" expression	output(1: "PUT") → "PUT" expression	output_rule = tokenPUT >> expression;	output_rule = tokenPUT >> expression;	{LA_IS, {T_OUTPUT_0}, { "output_rule", {\ {LA_IS, {""}, 2, { T_OUTPUT_0, "expression" }}\} }}\}
statement = labeled_point expression_or_cond_block__with_optional_assign goto_label input_rule output_rule ";;";	statement = labeled_point expression_or_cond_block__with_optional_assign goto_label input_rule output_rule ";;";	statement → labeled_point statement → expression_or_cond_block__with_optional_assign statement → goto_label statement → input_rule statement → output_rule statement → ";"	statement(1: " " , ";") → labeled_point statement(1: " " , ";" , 2: !" ") → expression_or_cond_block__with_optional_assign statement(1: "(" , "NOT", "+", "-", "0", "1", "2", "3", "4", "5", "6", "7", "8", "9", "IF") → expression_or_cond_block__with_optional_assign statement(1: "GOTO") → goto_label statement(1: "GET") → input_rule statement(1: "PUT") → output_rule statement(1: ";") → ";"	statement = labeled_point /* labeled_point first*/ expression_or_cond_block__with_optional_assign goto_label input_rule output_rule tokenSEMICOLON;	statement = labeled_point /* labeled_point first*/ expression_or_cond_block__with_optional_assign goto_label input_rule output_rule tokenSEMICOLON;	{LA_IS, { " ident_terminal " }, { "statement", {\ {LA_IS, {T_COLON_0}, 1, {"labeled_point"}}\} {LA_NOT, {T_COLON_0}, 1, {"expression_or_cond_block__with_optional_assign"}}\} }}\} {LA_IS, { "(" , T_NOT_0, "unsigned_value_terminal", T_ADD_0, T_SUB_0, T_IF_0 }, { "statement", {\ {LA_IS, {""}, 1, {"expression_or_cond_block__with_optional_assign"}}\} }}\} {LA_IS, {T_GOTO_0}, { "statement", {\ {LA_IS, {""}, 1, {"goto_label"}}\} }}\} {LA_IS, {T_INPUT_0}, { "statement", {\ {LA_IS, {""}, 1, {"input_rule"}}\} }}\} {LA_IS, {T_OUTPUT_0}, { "statement", {\ {LA_IS, {""}, 1, {"output_rule"}}\} }}\} {LA_IS, {T_SEMICOLON_0}, { "statement", {\ {LA_IS, {""}, 1, {";"}}\} }}\}
	statement__iteration = statement, statement__iteration ε;	statement__iteration → statement statement__iteration statement__iteration → ε	statement__iteration(1: " " , "(" , "NOT", "0", "1", "2", "3", "4", "5", "6", "7", "8", "9", "+", "-", "IF", "GOTO", "GET", "PUT", ";", ")) → statement statement__iteration statement__iteration(1: !" " , "(" , !"NOT", !"0", !"1", !"2", !"3", !"4", !"5", !"6", !"7", !"8", !"9", !" +", !" -", !"IF", !"GOTO", !"GET", !"PUT", !" , ")) → ε		statement__iteration = statement >> statement__iteration "";	{LA_IS, { " ident_terminal " , "(" , T_NOT_0, " unsigned_value_terminal ", T_ADD_0, T_SUB_0, T_IF_0, T_GOTO_0, T_INPUT_0, T_OUTPUT_0, T_SEMICOLON_0 }, { "statement__iteration", {\ {LA_IS, {""}, 2, { "statement", "statement__iteration" }}\} }}\} {LA_NOT, { " ident_terminal " , "(" , T_NOT_0, " unsigned_value_terminal ", T_ADD_0, T_SUB_0, T_IF_0, T_GOTO_0, T_INPUT_0, T_OUTPUT_0, T_SEMICOLON_0 }, { "statement__iteration", {\ {LA_IS, {""}, 0, { "" }}\} }}\}
block_statements = "(" , {statement} , ")" ;	block_statements = "(" , statement__iteration , ")";	block_statements → "(" statement__iteration ")"	block_statements(1: "(") → "(" statement__iteration ")"	block_statements = tokenBEGINBLOCK >> *statement >> tokenENDBLOCK;	block_statements = tokenBEGINBLOCK >> statement__iteration >> tokenENDBLOCK;	{LA_IS, {T_BEGIN_BLOCK_0}, { "block_statements", {\ {LA_IS, {""}, 3, {T_BEGIN_BLOCK_0, "statement__iteration", T_END_BLOCK_0 }}\} }}\}
	expression__optional = expression "";	expression__optional → expression expression__optional → ε	expression__optional(1: "(" , "NOT", "+", "-", " " , "0", "1", "2", "3", "4", "5", "6", "7", "8", "9", "IF") → expression expression__optional(1: !" "(" , !"NOT", !" +", !" -", !"0", !"1", !"2", !"3", !"4", !"5", !"6", !"7", !"8", !"9", !"IF") → ε		expression__optional = expression "";	{LA_IS, { "(" , T_NOT_0, T_ADD_0, T_SUB_0, "ident_terminal", "unsigned_value_terminal", T_IF_0 }, { "expression__optional", {\ {LA_IS, {""}, 1, { "expression" }}\} }}\} {LA_NOT, { "(" , T_NOT_0, T_ADD_0, T_SUB_0, "ident_terminal", "unsigned_value_terminal", T_IF_0 }, { "expression__optional", {\ {LA_IS, {""}, 0, { "" }}\} }}\}
program_rule = "NAME" , program_name , " , " , "BODY" , "DATA" , [declaration] , " , " , {statement} , "END";	program_rule = "NAME" , program_name , " , " , "BODY" , "DATA" , declaration__optional , " , " , statement__iteration , "END";	program_rule → "NAME" program_name " , " , "BODY" "DATA" declaration__optional " , " , statement__iteration "END"	program_rule(1: "NAME") → "NAME" program_name " , " , "BODY" "DATA" declaration__optional " , " , statement__iteration "END"	program_rule = BOUNDARIES >> tokenNAME >> program_name >> tokenSEMICOLON >> tokenBODY >> tokenDATA >> (- declaration) >> tokenSEMICOLON >> *statement >> tokenEND;	program_rule = BOUNDARIES >> tokenNAME >> program_name >> tokenSEMICOLON >> tokenBODY >> tokenDATA >> declaration__optional >> tokenSEMICOLON >> statement__iteration >> tokenEND;	{LA_IS, {T_NAME_0}, { "program_rule", {\ {LA_IS, {""}, 9, { T_NAME_0, "program_name", T_SEMICOLON_0, T_BODY_0, T_DATA_0, "declaration__optional", T_SEMICOLON_0, "statement__iteration", T_END_0 }}\} }}\}
	declaration__optional = declaration "";	declaration__optional → declaration declaration__optional → ε	declaration__optional(1: "INTEGER16") → declaration declaration__optional(1: !"INTEGER16") → ε		declaration__optional = declaration "";	{LA_IS, {T_DATA_TYPE_0}, { "declaration__optional", {\ {LA_IS, {""}, 1, { "declaration" }}\} }}\} {LA_NOT, {T_DATA_TYPE_0}, { "declaration__optional", {\ {LA_IS, {""}, 0, { "" }}\} }}\}
value = [sign] , unsigned_value;	value = sign__optional, unsigned_value;	value → sign__optional unsigned_value	value(1: "0", "1", "2", "3", "4", "5", "6", "7", "8", "9", "+", "-") → sign__optional unsigned_value	value = -sign >> unsigned_value >> BOUNDARIES;	value = sign__optional >> unsigned_value >> BOUNDARIES;	{LA_IS, { "unsigned_value_terminal", T_ADD_0, T_SUB_0 }, { "value", {\ {LA_IS, { "" }, 2, { "sign__optional", "unsigned_value" }}\} }}\}
	sign__optional = sign ε;	sign__optional → sign sign__optional → ε	sign__optional(1: "+" , "-") → sign sign__optional(1: !" +", !" -") → ε		sign__optional = sign "";	{LA_IS, {T_ADD_0, T_SUB_0}, { "sign__optional", {\ {LA_IS, { "" }, 1, { "sign" }}\} }}\} {LA_NOT, {T_ADD_0, T_SUB_0}, { "sign__optional", {\ {LA_IS, { "" }, 0, { "" }}\} }}\}
sign = sign_plus sign_minus;	sign = sign_plus sign_minus;	sign → sign_plus sign → sign_minus	sign(1: "+") → sign_plus sign(1: "-") → sign_minus	sign = sign_plus sign_minus;	sign = sign_plus sign_minus;	{LA_IS, {T_ADD_0}, { "sign", {\ {LA_IS, {""}, 1, { "sign_plus" }}\} }}\} {LA_IS, {T_SUB_0}, { "sign", {\ {LA_IS, {""}, 1, { "sign_minus" }}\}

				BOUNDARY__LINE_FEED = "\n"; BOUNDARY__NULL = "\0"; NO_BOUNDARY = ""; #define WHITESPACES \ STRICT_BOUNDARIES, \ BOUNDARIES, \ BOUNDARY, \ BOUNDARY__SPACE, \ BOUNDARY__TAB, \ BOUNDARY__VERTICAL_TAB, \ BOUNDARY__FORM_FEED, \ BOUNDARY__CARRIAGE_RETURN, \ BOUNDARY__LINE_FEED, \ BOUNDARY__NULL, \ NO_BOUNDARY	BOUNDARY__CARRIAGE_RETURN = "\r"; BOUNDARY__LINE_FEED = "\n"; BOUNDARY__NULL = "\0"; NO_BOUNDARY = ""; #define WHITESPACES \ STRICT_BOUNDARIES, \ BOUNDARIES, \ BOUNDARY, \ BOUNDARY__SPACE, \ BOUNDARY__TAB, \ BOUNDARY__VERTICAL_TAB, \ BOUNDARY__FORM_FEED, \ BOUNDARY__CARRIAGE_RETURN, \ BOUNDARY__LINE_FEED, \ BOUNDARY__NULL, \ NO_BOUNDARY	
						\ \ \ { LA_IS, { T_NAME_0 }, { "program____part1", \ { LA_IS, {""}, 7, { T_NAME_0, "program_name", T_SEMICOLON_0, T_BODY_0, T_DATA_0, "declaration_optional", T_SEMICOLON_0 }} \ } } \ } \ }, \ "program_rule"